

PRANAV NAIR

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SUMMARY

Innovative AI Engineer with expertise in developing intelligent, production-grade agents using RAG, LangChain, and LangGraph. Experienced in multi-agent systems, LLM fine-tuning, and full-stack development with modern web technologies. Driven by a passion for designing scalable, high-performance solutions that turn complex challenges into practical, AI-powered products.

SKILLS

- **Generative AI:** RAG Development & Deployment, LangChain, LangGraph, Vercel AI SDK, Agno AI Implementation, Multi-Agent System Architecture, Model Fine-Tuning, Optimization
- **Front-End:** Next.js, Three.js, TailwindCSS, Blender, After Effects
- **Back-End:** Node.js, Express, Python
- **Databases:** MongoDB, MySQL, Pinecone, Chroma
- **Advanced Concepts:** Advanced Machine Learning Concepts and Fine tuning Techniques, Scalable AI Agent Deployment

PROJECTS

Multi-Agent Customer Support Assistant

LangChain, SQL Agents, LLM Orchestration

- Designed and deployed a **multi-agent architecture** that cut average support response time by 50% through intelligent task delegation and context sharing.
- Integrated **SQL-powered retrieval agents** for dynamic database queries, enhancing response precision and factual accuracy by 30%.

Large Language Model Alignment Study

Python, PyTorch, Hugging Face, QLoRA, DPO

- Conducted fine-tuning of the **Llama-3-8B** model using **QLoRA** and **DPO** on the *Anthropic HH-RLHF* dataset to align model behavior with human preferences.
- Achieved a **44.3% reduction in toxicity** and **7.9% increase in helpfulness** through QLoRA, while DPO provided balanced safety alignment under limited GPU resources.

AI-Powered Knowledge Retrieval System

LangChain, LangGraph, Vector Databases

- Engineered an optimized **RAG pipeline** using LangGraph to reduce query latency by 35% and boost retrieval precision across complex knowledge bases.
- Designed **efficient vector indexing and chunking strategies**, cutting large-scale data retrieval times by 40% while maintaining contextual relevance.

EXPERIENCE

AI Full-Stack Developer, Scogo Networks – Mumbai, India

Dec 2024 – Jun 2025

- Developed and integrated **RAG pipelines for AI voice agents**, enabling context-aware query handling and improving response accuracy.
- Contributed across the full stack—from front-end workflows to back-end logic—ensuring seamless AI integration and optimized system performance.

Software Developer Intern, HashedBit Innovations – Mumbai, India

Apr–Sep 2024

- Revamped and optimized full-stack architecture, improving load efficiency by 15% and ensuring smoother API-client communication.
- Built and deployed scalable **RESTful APIs** with Node.js and Express, integrating CI/CD pipelines for production releases handling 10,000+ daily requests.

EDUCATION

AI Engineering, MisogiAI by Masai, Bangalore

Jun 2025 – Present

Bachelor of Technology: Computer Engineering (AI/ML), Pillai College of Engineering, Navi Mumbai

Aug 2021 – May 2025